

Determination of Susceptibility of Coal to spontaneous combustion using Crossing Point Temperature method

1) Define Spontaneous Combustion. (Spontaneous combustion or spontaneous ignition is a type of combustion which occurs by the process of self-heating (increase in temperature due to exothermic internal reactions) of coal or other carbonaceous material, followed by thermal runaway (self-heating which rapidly accelerates to high temperatures) and finally, ignition.)

2) What do you understand by CPT? (Crossing point temperature (CPT) is the temperature at which the coal temperature coincides with that of the furnace temperature or bath temperature in °C. CPT or critical oxidation temperature gives an idea about the proneness of coal to auto-oxidation or self-heating. The higher the crossing point temperature value, the less will be susceptibility of coal to spontaneous heating.)

3) What are the conditions which should be available for the occurrence of spontaneous combustion?

- Coal must be in a form which can oxidize readily at ambient temperature.
- Oxygen must be available to support the oxidation.
- Favourable conditions for accumulation of heat must be available.

4) Why do we use glass wool? (So that coal doesn't block the path of air and coal remains in the form of layer or seam resembling that in mines)

5) What is the significance of spiral tube? (Increase the air travel distance so that it could adapt the environment created by glycerine bath better)

6) What is the use of stirrer? (To have constant temperature throughout the glycerine bath)

7) What are the specifications of the experiment?

- 20 gram of coal sample of size -212 micron or -72 mesh size.
- Heating rate = 1°C per minute.
- O₂ flow rate = 80 ml per minute.

5) What are the limitations of the experiment? (The major limitation of CPT experiment is that the rate of flow of O₂, rate of heating, particle size of coal, etc. have to be properly maintained. Otherwise with slight change in any of the above conditions the CPT may change. This method is very time consuming and takes 3-4 hrs to complete experiment and failure of power supply even for a few minutes during experimentation especially when the temperature approaches CPT, gives erroneous results and the whole experiment is to be repeated.)

Determination of Inflammability Index of a given coal sample by Godbert-Greenwald apparatus

1) What is the basic idea of this experiment?

2) Mathematically define Explosibility factor or index. ($Ex = 100/(100-S)$, where S is percentage of stone dust at which flame propagation stopped)

3) How do you monitor the samples when it enters the heating silica tube? (Angled stainless steel used as mirror)

4) What is the material used making in sample powder injection module? (Brass)

5) How the dust sample reaches heating silica tube? (With the help of Air Input Assembly)

6) Which control module is called the heart of this experiment? (Heat Control Module)

7) What should be the pressure of the regulated air and Valve open time for this experiment? (Pressure = 5 psi and Valve open time = 0.25 seconds)

8) Which coal is safer the one with higher Explosibility factor or the one with lower Explosibility factor? (The one with lower Explosibility factor)

9) What is the lower flammability limit of coal in general? (30 gm per m³ to 300 gm per m³)

Determination of airborne respirable dust concentration using Gravimetric Dust Sampler (Model-MRE 113A) and Konimeter

A) Gravimetric Dust Sampler

- 1) What is the formula to calculate maximum allowable dust concentration according to the Indian mines regulation 2017? ((10/ percentage silica dust) mg per m³)
- 2) What is the respirable dust particle size? (Less than 10 micro-meters)
- 3) At what level do we hang Gravimetric Dust Sampler in mine roadways? (2/3rd of the full height of mine roadway or nose level)
- 4) What is the operating time of Gravimetric Dust Sampler? (It operates for a complete shift which is in general of 8 hours in India)
- 5) What is the suction rate of Gravimetric Dust Sampler? (2.5 litre per minute)
- 6) What does the deviation of suction rate convey? (Greater than 2.5 litre per minute means overcharging and less than 2.5 litre per minute means discharged batter or filter paper with dust)
- 7) What is the specification of battery used in Gravimetric Dust Sampler? (7.5 V, 90 mA Ni-Cd rechargeable battery)
- 8) What is the type and dimension of the filter paper used? (55 mm diameter, circular, glass fibre type)
- 9) What is the significance of slotter? (Slotter or restrictor restricts most of the air dust particles which are above the size range of respirable dust particle size and also controls air cross-flow)
- 10) What is the significance of multi-channel horizontal elutriator? (The dust samples which are larger than the respirable dust size (10 micrometres) settles down due to their heavy weight and gravity and this multi-channel provides longer time to do so)
- 11) How do you take count of the air flow through Gravimetric Dust Sampler? (With the help of flowmeter connected with the Gravimetric Dust Sampler and we can also calculate the air volume flown with the help of time elapsed and flow rate)
- 12) What is the unit in which dust concentration is represented in Gravimetric Dust Sampler? (mg per m³)

B) Konimeter

- 1) What is the volume of air sucked in one time by Konimeter? (5 cc)
- 2) In how many stages dust deposition chamber is divided? (30)
- 3) How do we know 5-micron dust size in Konimeter? (Particles which were trapped in between the parallel lines have size approximately equal to 5 microns)
- 4) What is the reference size of dust particles in Konimeter? (5 microns)
- 5) How many sectors are seen in the microscope of the Konimeter? (2)
- 6) What is the angle of each sector seen in the microscope of the Konimeter? (18°)
- 7) What is the output unit of Konimeter? (ppcc)
- 8) What are the limitations of Konimeter?
 - The performance or efficiency under varying conditions is unpredictable. Different types of Konimeter gives different results.
 - It is difficult to evaluate a heavy deposit with high dust concentration as in blasting.
 - The quantity of dust deposited depends on the nature and thickness of adhesive (petroleum jelly) film.
 - It has a tendency to break up aggregates due to high impact velocity.
 - Only spot sampling is possible.

Determination of airborne respirable dust concentration using Personal Dust Sampler CIP 10

- 1) What are the specifications of Personal Dust Sampler? (Case dimension - $100 * 70 * 25$ mm and weight is 300 grams)
- 2) How sample is taken in case of Personal Dust Sampler? (Sample air is induced into the sampler by a sponge ring contained in a rotating plastic cup, which rotates at a speed of approximately 7000 rpm ((6800 $\pm$ 200) rpm))
- 3) What is the sampling flow rate of Personal Dust Sampler? (10 litre per minute (the reason why it is called CIP 10), it is controlled by the rotation rate of sponge ring)
- 4) Output unit of Personal Dust Sampler? (mg per m^3)
- 5) What are the different pore sizes and thicknesses in which foam sponges are available? (Pore sizes - 45 (coarse), 60 (medium), 100 (fine) per inch and thickness - 10, 20 mm; fine one is available in 10 mm thickness only)
- 6) What is the significance of different pore sizes of foam sponges? (Coarser one allows fine particles to move forward but restricts big particles and the fine pored sponge collects the fine or respirable dust particles)

Determination of airborne respirable dust concentration using Real-time Aerosol Monitor (Model: RAM-1)

- 1) What is the allowable limit of dust according to CMR 2017? (2mg per m^3 or maximum 5 % free silica)
- 2) The sensors used in RAM-1 are of which spectral region? (EM radiation in the near-infrared region)
- 3) What is the Total and Purge or False flow in case of RAM-1? (Total - 2.2 litre per minute (sampling flow 2.0 litre per minute and 10% Purge flow) and False or Purge flow - 0.2 litre per minute, this purge flow provides fresh air for continuous flushing of fresh air)
- 4) What are the different time constants in RAM-1?
 - 0.5 second
 - 2.0 second
 - 8.0 second
 - 32.0 second
- 5) Should we keep the time constant low or high? (It depends on the situation, if we have ample amount of time, we should go for higher time constant as more time will decrease the fluctuations of reading and another case is when you need immediate reading, in this case lower time constant should be selected)
- 6) What are the different ranges in which respirable dust concentration is measured in Real-time Aerosol Monitor (Model: RAM-1)?
 - i) 0-2 mg per m^3
 - ii) 2-20 mg per m^3
 - iii) 20-200 mg per m^3
- 7) Why we don't directly use higher range? (If we directly use higher range, it may cause problems in the sensors used)